
PrimeCore Minerals

Wi-Fi Design

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Introduction

Employees today expect some flexibility in where they work and this could be at home, in the office or presenting in meetings. To achieve this goal, laptops are the main hardware supplied to employees and the easiest way to keep employees with laptops on the network is via Wi-Fi. By enabling and configuring the Wi-Fi correctly, another SSID can be configured for visitors/contractors to use while in the office or at site.

As Wi-Fi can't be setup in a virtual environment, VLANs were used to demonstrate the segregation of the network so firewall rules could be implemented and demonstrated.

Configuration

Two different SSIDs will be setup so employees could connect to the PCM-Employee SSID, and visitors/contractors could connect to the PCM-Visitors SSID.

The features for the PCM-Employee network include:

- SSID called PCM-Employee
- Firewall rules updated to monitor and permit/deny access to this SSID
- Wi-Fi 7 (802.11be) with Multi-Link Operation (MLO) plus backward compatible to include Wi-Fi 6 and Wi-Fi 5
- Join the 2.4GHz, 5GHz and 6GHz channels and recommend connections connect to the highest band
- WPA3-Enterprise security
- EAP-TLS (to allow Wi-Fi certification for company devices)
- AAA/RADIUS Server (to allow device certification authentication)
- Client-Isolation (so employees can't connect to other employee devices)
- Fast Roaming (802.11 r/k/v) (allows employees to walk around the office or site and connect to the AP with the strongest connection)

The features for the PCM-Visitor network include

- SSID called PCM-Visitor
- Firewall rules updated to monitor and permit/deny access to this SSID
- Wi-Fi 6 and Wi-Fi 5 enabled
- Wi-Fi 7 disabled (avoids congestion on the Wi-Fi 7 band)
- SSID enabled as Guest SSID in Meraki dashboard
- Open network with Web Authentication

- o Web Authentication approved via Captive Portal (Splash Page)
 - Captive Portal request access from a valid PCM employee, who sets end date for visitor.
 - An example of the captive portal is in figure 1¹
- Client Isolation (so they can't see any other device on the network)
- Fast Roaming (802.11 r/k/v) (allows visitors to walk around the office or site and connect to the AP with the strongest connection)

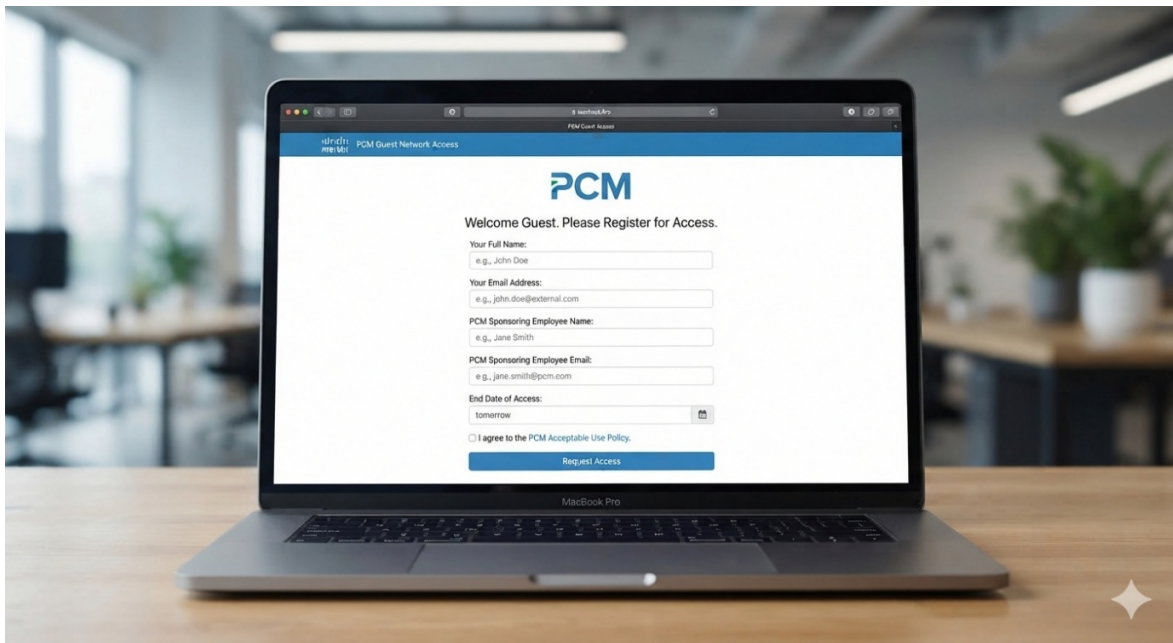


Figure 1 – example of a captive portal the company could create so visitors/contractors obtain access to PCM_Visitor Wi-Fi

Setup and Configuration

Cisco has setup the following website which steps through setting up a Meraki Dashboard (including for companies for the first time) then goes through how to add the AP to your hardware list and how to configure the features mentioned above and this is the best way to setup the APs securely and consistently.

https://documentation.meraki.com/Wireless/Install_and_Get_Started/Quick_Start_Guides

Reference

1. Google AI

Request: create a generic image of a meraki captive portal where it asks a user to enter name and email address of PCM employee as well as the access end date

Result: image in figure1